

PAPER_308 — Spiral Arm Torque Gravitational Amplifier

$\tau_{\text{spiral}}(10 \text{ Gyr}) = 2.046 \mid T_{\text{pattern}} = 307 \text{ Myr} \mid d\tau/dt = 2.741 \times H_0_{\text{SH0ES}}$

Session 88 | 30th C++ UQFF module | FIRST Spiral+SN Ia UQFF 2.0

Module: SPIRAL_SUPERNOVAE_UQFF_MODULE.cpp

Classification: FIRST UQFF spiral arm pattern speed gravitational amplifier

Status: Unique physics — no prior UQFF spiral torque gravity amplification study

Abstract

Spiral galaxies evolve under pattern-speed-driven star-formation torques whose cumulative gravitational amplitudes differ fundamentally from static NFW or Keplerian assumptions. Within the UQFF (Unified Quantum Field Framework) 2.0 pipeline, a dimensionless spiral torque factor $\tau_{\text{spiral}} = (M_{\text{gas}}/M) \times \Omega_p \times t$ accumulates over cosmic time and multiplies the base gravitational term. At 10 Gyr the amplifier reaches $\tau = 2.046$, boosting effective gravity by a factor of $3.046\times$. The torque rate $d\tau/dt = 6.483 \times 10^{-18} \text{ s}^{-1}$ exceeds H_0_{SH0ES} ($73.0 \text{ km/s/Mpc} = 2.366 \times 10^{-18} \text{ s}^{-1}$) by a factor of 2.741, establishing a new UQFF cosmic-rate comparison.

2. System Parameters

Parameter	Value	Notes
M (galaxy)	$1.989 \times 10^{41} \text{ kg}$	$1 \times 10^{11} M_{\text{sun}}$ (Milky-Way class)
M_{gas} (arm gas)	$1.989 \times 10^{39} \text{ kg}$	$1 \times 10^9 M_{\text{sun}}$
r	$9.258 \times 10^{20} \text{ m}$	$\sim 30 \text{ kpc}$ half-radius
Ω_p (pattern speed)	$6.483 \times 10^{-16} \text{ rad/s}$	20 km/s/kpc
H_0_{SH0ES}	73.0 km/s/Mpc	SH0ES, Riess et al. 2022
$f_{\text{gas}} = M_{\text{gas}}/M$	0.01	Gas arm mass fraction

3. Derivation

3.1 Dimensionless Spiral Torque

The spiral arm torque amplifier in UQFF 2.0 is defined as a dimensionless running accumulation of pattern momentum:

$$\tau_{\text{spiral}}(t) = \frac{M_{\text{gas}}}{M} \cdot \Omega_p \cdot t$$

- $f_{\text{gas}} = 10^9 / 10^{11} = \mathbf{0.01}$
- $\Omega_p = 20.0 \times 10^3 / 3.086 \times 10^{19} = \mathbf{6.483 \times 10^{-16} \text{ rad/s}}$
- $t (10 \text{ Gyr}) = 10 \times 10^9 \times 3.15576 \times 10^7 = \mathbf{3.156 \times 10^{17} \text{ s}}$

$$\tau_{\text{spiral}}(10 \text{ Gyr}) = 0.01 \times 6.483 \times 10^{-16} \times 3.156 \times 10^{17} = \boxed{2.046}$$

3.2 Gravity Amplification

The UQFF 2.0 pipeline applies τ as a multiplicative stage:

$$g_{\text{pipeline}}(t) = \frac{GM}{r^2} \cdot \underbrace{(1 + H_z t)}_{\text{Hubble}} \cdot \underbrace{(1 + \tau_{\text{spiral}})}_{\text{torque}} \cdot \underbrace{(1 - B/B_{\text{crit}})}_{\text{SC}} \cdot \underbrace{(1 + f_{\text{TRZ}})}_{\text{TRZ}}$$

At $t = 10 \text{ Gyr}$, $z = 0$:

$$g_{\text{amp}} = 1 + \tau_{\text{spiral}} = 1 + 2.046 = \boxed{3.046}$$

Effective gravity is **3× stronger at 10 Gyr** than at formation, driven entirely by spiral arm pattern momentum accumulation.

3.3 Pattern Period

$$T_{\text{pattern}} = \frac{2\pi}{\Omega_p} = \frac{2\pi}{6.483 \times 10^{-16}} = 9.692 \times 10^{15} \text{ s} = \boxed{307 \text{ Myr}}$$

A spiral arm completes one full rotation pattern in 307 Myr — consistent with observed grand-design spiral arm lifetimes.

3.4 Torque Rate vs Hubble Rate

$$\frac{d\tau}{dt} = f_{\text{gas}} \cdot \Omega_p = 6.483 \times 10^{-18} \text{ s}^{-1}$$

$$H_{0,\text{SH0ES}} = \frac{73.0 \times 10^3}{3.086 \times 10^{22}} = 2.366 \times 10^{-18} \text{ s}^{-1}$$

$$\frac{d\tau/dt}{H_0} = \frac{6.483 \times 10^{-18}}{2.366 \times 10^{-18}} = \boxed{2.741}$$

The spiral torque accumulation rate exceeds the Hubble expansion rate by 2.741×. This means spiral arm gravitational evolution is cosmologically faster than universal expansion, a key UQFF prediction distinguishing galactic internal dynamics from global metric expansion.

4. Physical Interpretation

The spiral arm torque factor captures how non-axisymmetric mass flows (gas infall, density wave sweeping, star formation) cumulatively torque the gravitational potential. In UQFF 2.0, this is written as a dimensionless running factor on the total gravity pipeline. The key results:

- **$\tau_{\text{spiral}}(10 \text{ Gyr}) = 2.046$** — Spiral galaxies experience $>3\times$ internal gravity enhancement relative to their formation epoch
- **$T_{\text{pattern}} = 307 \text{ Myr}$** — Sets the natural clock for spiral arm-driven enrichment cycles (OB association lifetimes, molecular cloud compression cycles)

- $d\tau/dt / H_0 = 2.741$ — Torque evolution rate 2.7× faster than cosmic expansion: internal galactic structure evolves more rapidly than expanding-universe metrics suggest

5. UQFF 2.0 Equations (Full Pipeline, $t = 10$ Gyr)

Term	Value	Notes
$g_{\text{base}} = GM/r^2$	$1.549 \times 10^{-11} \text{ m/s}^2$	Reference gravity at 30 kpc
Hubble factor ($1 + H_0 \cdot t$)	system-z dependent	Expansion correction
Torque factor ($1 + \tau$)	3.046	PAPER_308 key result at 10 Gyr
SC factor ($1 - B/B_{\text{crit}}$)	≈ 1.0	Galactic $B \ll B_{\text{crit}}$
TRZ factor ($1 + f_{\text{TRZ}}$)	1.10	Resonance zone correction
Ug_{sum}	$2 \times g_{\text{base}}$	Magnetic-dipole + vacuum-SC
Λ term	~ 0	Sub-dominant at galactic scales
Quantum term	$\hbar/(m_H \cdot \Delta x^2)$	HUP contribution
g_{DM}	$1.316 \times 10^{-11} \text{ m/s}^2$	PAPER_310 dark matter partition
a_{SN}	$3.096 \times 10^5 \text{ m/s}^2$	PAPER_309 SN Ia radiation pressure

6. Novel Findings (UQFF Firsts)

- **FIRST** UQFF spiral arm pattern speed gravitational amplifier
- **FIRST** UQFF dimensionless torque running factor in 9-term pipeline
- **FIRST** UQFF galactic-scale $d\tau/dt$ vs H_0 comparative rate analysis
- $\tau = 2.046$ exceeds unity: spiral arm evolution **non-perturbatively** amplifies gravity at late cosmic time

7. References

- Riess et al. 2022, ApJ 934 L7 (SH0ES $H_0 = 73.04 \text{ km/s/Mpc}$)
- Binney & Tremaine, *Galactic Dynamics* 2nd ed. (spiral arm pattern speeds)
- UQFF 2.0 Architecture — ARCHITECTURE_FLOW_DIAGRAM.md v4.4.0 CANONICAL
- MAIN_1_CoAnQi.cpp SOURCE4 (lines 25623-26026) — UQFF/MUGE framework
- Session 88 — SPIRAL_SUPERNOVAE_UQFF_MODULE.cpp (UQFF 2.0 implementation)